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Project 2 :
Analysis of Teenage Mental Health:
Profiling Digital Habits and Well-Being Using K-Means and DBSCAN
An Unsupervised Learning Approach (Clustering)

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1.0 Introduction

1.1 Introduction to the project

In this era of digital technology, there is a massive increase in screen time and social media usage among teenagers that has triggered global issues and conversation about mental health among youth. Through this project, it will investigate how digital lifestyle, physical habits, sleep schedule and psychological signs have interacted within the youth population. By evaluating these factors and connection, this project aims to identify the distinctness of youth behaviours and well being.

1.2 Brief Explanation of clustering analysis in general

Clustering is an unsupervised machine learning technique about partitioning a set of data objects into subsets called clusters. It is done by grouping the data based on their features that have similarities without relying on the pre-existing target labels. Additionally, clustering algorithms maximize the inter-cluster distance and minimize the intra-cluster distance that can help to discover the underlying structure or the natural grouping in the data.

1.3 Project Objective

The primary objective of this project is to apply and compare the clustering algorithms which are K-means and Density-based clustering on the dataset about teenage mental health. By using the Principal Component Analysis (PCA), the project attempts to group determine the data to see clearly the different types of habits and behaviours among the youth.

1.4 Purpose of using the selected dataset

The dataset “Teen_Mental_Health_Dataset.csv” is suited for this project because it combines both qualitative and quantitative features alongside the mental health outcome. It can provide an empirical foundation to test how digital overexposure can cause mental health risk among the youth.

2.0 Business and Data Understanding

2.1 Problem formulation for the selected dataset

Mental treatment by schools and clinical professionals are often reactive or oversimplified. Schools and doctors always wait until the teenager is already struggling before they help or use a one-size-fits-all approach. Therefore this project solves that problem by looking at the teen's daily habits and mental health data together to find common group profiles without biasing the model with conclusive diagnosis labels. This will translate into identifying whether certain combinations of screen time and poor sleep consistently group with high anxiety or addiction metrics.

2.2 Overview of Dataset

The Teen_Mental_Health_Dataset.csv contains sample size of 1200 individual student records that has been distributed across the 13 attributes that track a comprehensive mix of teenage demographics such as **age** and **gender**, digital lifestyle like **daily_social_media_hours**, **platform_usage** and **screen_time_before_sleep**, the physical routines such as **sleep_hours** and **physical_activity** and lastly about the psychological health such as **social_interaction_level**, **academic_performance**, **stress_level**, **anxiety_level**, **addiction_level** and **depression_label**. The dataset has been immediately going through preprocessing process to clean the zero missing value, null or NaN values all across the row and column to ensure the unsupervised feature selection and clustering method can be done well.

2.3 Importance of dataset to the project

This dataset acts as a primary evidence base for the unsupervised model because it records many different features for the exact 1200 individuals where it can allow clustering algorithms to evaluate the correlation and catch the pattern.

2.4 Description of important attributes or features

Table 1 explains the description of the important attributes selected to develop different groups and clusters of the students.

Table 1: Overview of the Dataset

Attributes	Data Type	Description
age	Ratio	Basics background of the student
gender	Nominal	
daily_social_media_hours	Ratio	Metrics tracking for the duration
screen_time_before_sleep	Ratio	

platform_usage	Nominal	Preferred social media used such as Instagram, Facebook or TikTok
sleep_hours	Ratio	Core daily life routines such as rest, physical exercise and real world socialization levels
physical_activity	Ratio	
social_interaction_level	Ordinal	
academic_performance	Ratio	Metrics that can measure the school outcomes alongside with the physical state and medical symptoms
stress_level	Ratio	
anxiety_level	Ratio	
addiction_level	Ratio	
depression_label	Nominal (Binary)	

2.5 Expected business value or insight from the analysis

This analysis can help school and health organizations to create profiles of different types of teenagers. Instead of using the same general advice for everyone, they can target a specific group such as teenagers who spend much time on social media, rarely exercise and feel highly anxious. This can make the school or health organizations send specific helpful tips or counselor direct support to the teenagers who need it the most.

3.0 Data Preparation and Preprocessing

3.1 Overview of the data preparation

Data preparation is the process of cleaning and transforming raw data into a suitable format for processing and analysis. It is an essential step in data mining because raw data may contain inconsistencies, missing values, duplicate records, and other errors that can affect the quality of the results. Therefore, data preparation helps ensure that the dataset is accurate, complete, and ready for further analysis.

The importance of data preparation lies in its ability to identify and correct errors before analysis is conducted. Detecting errors at an early stage reduces the risk of inaccurate results and minimizes the difficulty of correcting issues later in the process. In addition, cleaning and transforming the data help produce a high-quality dataset that is consistent, complete, and reliable for analysis.

Furthermore, properly prepared data improves the performance of data mining and machine learning models by reducing noise and bias within the dataset. High-quality data enables more accurate predictions, insights, and reliable outcomes. As a result, data preparation contributes to effective decision-making and the overall success of a data mining project.

3.2 Explanation of every preprocessing step involved

In this project, several data preparation steps were performed to ensure the dataset was suitable for data mining analysis. The data preparation process began by importing the Teen_Mental_Health_Dataset.csv dataset into Python using the Pandas library. Exploratory data analysis(EDA) was conducted to understand the structure and quality of the dataset. This includes checking the dataset information(df.info()), accessing original dimensions(df.shape) and detecting duplicate records that could be biased to other clusters. Value distributions for categorical variables were reviewed to ensure consistency, such as trimming white spaces and aligning text cases for social_interaction level and platform_usage.

Several preprocessing techniques were implemented in this project to refine and clean data for substance-based clustering algorithms. Missing values were handled through average(mean) of the column to keep data filled and complete. Next, we converted text categories into numbers where we use binary mapping for gender , ordinal encoding for interaction levels and One-Hot encoding for platform names. Apart from that, StandardScaler was used to normalize the data. It ensures that features with large numbers do not overwhelm or affect features with small numbers. Finally, depression_label was removed making the model unsupervised and used Principal Component Analysis(PCA) to simplify the data into two main components which makes it easier for clustering analysis

3.3 Handling of missing values

Clustering algorithms require a complete dataset to perform distance calculations. Based on figure 3.3.1 missing values in numerical columns were handled using mean imputation fillna(x.mean()) which replaces gaps with average of their respective columns . Then, missing values were checked using isnull().sum() to confirm whether missing values still exist. This approach was selected to preserve the overall sample size while maintaining the mathematical continuity required for centroid-based clustering.

```
X = X.fillna(X.mean(numeric_only=True))
missing_after = X.isnull().sum().sum()
print(f" -> Missing Value Report (AFTER CLEANING): Found {missing_after} missing data cells.")
```

Figure 3.3.1: Snippet code for handling missing value

3.4 Handling of duplicate data

Based on figure 3.4.1, Duplicate verification was performed to ensure that the integrity of the density-based and partitioning models was maintained. Dataset was first checked if any duplicates exist using duplicated.sum(). Function df.drop_duplicates() will then remove any redundant rows when detected. This step is critical in unsupervised learning to prevent the artificial inflation of data density, which would bias the cluster formation and lead to skewness in model result.

```
duplicates = df.duplicated().sum()
print(f" -> Duplicate check: Found {duplicates} duplicate rows.")
if duplicates > 0:
    df.drop_duplicates(inplace=True)
    print("    Justification: Duplicates removed to prevent artificial density cluster bias.")
```

Figure 3.4.1: Snippet code for Handling Duplicate Data

3.5 Encoding of categorical variables

Based on figure 3.5.1, Three specific encoding techniques were applied namely, Binary Mapping, Ordinal Encoding and One-Hot Encoding. Binary mapping was used to change categorical variables such as gender into numerical values (0 and 1) with map() function. Ordinal Encoding on the other hand was used to map social_interaction_level to an ordinal scale (0, 1, 2) to resemble the inherent logical hierarchy (Low, Medium, High) with map() function. Lastly, One-Hot Encoding was used to transform platform_usage variables into multiple binary columns with the pd.get_dummies() function, which turns each category into a separate column and marks rows with 1 if they belong to that category and 0 if not. It is to prevent the model from assigning an artificial mathematical hierarchy to nominal categories.

```
# Binary mapping
df['gender'] = df['gender'].map({'male': 0, 'female': 1})

# Ordinal encoding
df['social_interaction_level'] = df['social_interaction_level'].map({'low': 0, 'medium': 1, 'high': 2})

# One-Hot encoding
platform_dummies = pd.get_dummies(df['platform_usage'], prefix='platform', dtype=int)
df = pd.concat([df, platform_dummies], axis=1)
df.drop(columns=['platform_usage'], inplace=True)
print(" -> Encoding completed (Binary Mapping, Ordinal Scale, One-Hot Multi-class).")
```

Figure 3.5.1: Snippet code for Encoding of Categorical Value

3.6 Feature Scaling and Normalization of numerical data

Since K-Means and DBSCAN rely on Euclidean distance, feature scaling is necessary. Based on figure 3.6.1, The dataset undergoes standardization using StandardScaler, which transforms features to a distribution with a mean 0 and standard deviation 1. This ensures that variables with larger ranges do not dominate distance and allow those with smaller ranges to contribute fairly to cluster formation.

```
scaler = StandardScaler()
X_scaled = scaler.fit_transform(X)
print(" -> Feature Scaling: Standardization completed (Mean=0, Variance=1).")
```

Figure 3.6.1: Snippet code for Feature Scaling and Normalization

3.7 Text Normalization & Standardisation

Based on figure 3.7.1, The code performs text normalization and standardization on categorical columns to ensure consistency before encoding or clustering. In the social_interaction_level column, leading and trailing spaces were removed using str.strip() function while str.lower() converts text to lowercase ensuring consistency. In the platform_usage column, extra spaces and inconsistent naming were fixed and clean using str.strip() function. Finally, the print() statement confirms that the process is complete, meaning the dataset now has clean and standardized text values are ready for further preprocessing

```
df['social_interaction_level'] = df['social_interaction_level'].str.strip().str.lower()
df['platform_usage'] = df['platform_usage'].str.strip().replace({'Tik Tok': 'TikTok'})
print(" -> Text standardization completed (white spaces trimmed, case alignment).")
```

Figure 3.7.1: Snippet code for Text Normalization & Standardisation

3.8 Feature selection

Based on figure 3.8.1, the depression_label was removed from the feature set. Including the target label in an unsupervised pipeline can cause "target leakage," where the model could unintentionally use the label to form clusters. By isolating only behavioral and usage features, it ensures the model form natural, independent patterns cluster within the student data.

```
if 'depression_label' in df.columns:
    X = df.drop(columns=['depression_label'])
    print(" -> Feature Selection: Removed 'depression_label' target column.")
else:
    X = df.copy()
```

Figure 3.8.1: Snippet code for encoding of categorical value

3.9 Sample input and output for each important preprocessing step

Based on figure 3.8.1, output shows that the dataset was cleaned and prepared before clustering. Missing values and duplicates were checked and found to be zero. Text was standardized for consistency, and categorical variables were encoded using binary, ordinal, and one-hot methods. The target label was removed to avoid leakage and features were scaled to balance ranges. Finally, PCA reduced 14 features into 2 components while 21.70% of the variance was preserved.

```
--------------------------------------------------
REQUIREMENT 1: DATA PREPROCESSING & DATA CLEANING
--------------------------------------------------
 -> Missing Value Report (BEFORE CLEANING): Found 0 missing data cells.
 -> Duplicate check: Found 0 duplicate rows.
 -> Text standardization completed (white spaces trimmed, case alignment).
 -> Encoding completed (Binary Mapping, Ordinal Scale, One-Hot Multi-class).
 -> Feature Selection: Removed 'depression_label' target column.
 -> Sampling Action: Preserved 100% of dataset records to maintain density structures.
 -> Missing Value Report (AFTER CLEANING): Found 0 missing data cells.
 -> Feature Scaling: Standardization completed (Mean=0, Variance=1).
 -> PCA Dimension Reduction: 14 features reduced to 2 components.
    Variance Preserved: 21.70%
```

Figure 3.9.1: Snippet code for encoding of categorical value

4.0 Model Development: Unsupervised Learning (Clustering)

4.1 Overview of the Selected Clustering Algorithm

This study implements two unsupervised clustering algorithms, namely K-Means and DBSCAN, to identify hidden patterns within the Teen Mental Health dataset. Unlike supervised learning, clustering does not require labelled target variables. Instead, it groups observations according to their similarity based on the selected features.

4.1.1 K-Means Clustering

K-Means is a distance-based partitioning algorithm that divides the dataset into K distinct clusters. The algorithm begins by selecting K initial centroids and repeatedly assigns each data point to its nearest centroid based on Euclidean distance. The centroid positions are then recalculated until the cluster assignments no longer change or convergence is reached. Since K-Means is distance-based, all numerical features were standardized before clustering to ensure that variables with larger numerical ranges did not dominate the distance calculations.

4.1.2 DBSCAN Clustering

DBSCAN is a density-based clustering algorithm that groups data points according to their local density rather than predefined cluster centres. Unlike K-Means, DBSCAN can discover clusters with irregular shapes and automatically identify outliers as noise points. The algorithm requires two parameters:

- eps : the maximum neighbourhood radius for a point to be considered a neighbour.
- min_samples : the minimum number of neighbouring points required to form a dense region

4.2 Justification for Selecting the Two Algorithms

Two clustering algorithms, namely K-Means and DBSCAN were selected for this study because they represent two fundamentally different clustering approaches and allow a comprehensive comparison of clustering performance on the Teen Mental Health dataset. K-Means is a partitioning-based clustering algorithm that is widely used due to its simplicity, computational efficiency, and ability to produce compact and well-separated clusters. It performs well when the data contains clusters with relatively spherical shapes and similar densities. Additionally, K-Means generates cluster centroids, making the clustering results easier to interpret and analyse.

In contrast, DBSCAN is a density-based clustering algorithm that groups data points according to their local density rather than assigning every observation to a predefined cluster. One of its major advantages is its ability to discover clusters with irregular shapes while simultaneously identifying outliers as noise points. Unlike K-Means, DBSCAN does not require the number of

clusters to be specified before training, making it suitable for datasets where the underlying cluster structure is unknown.

The combination of these two algorithms enables a meaningful comparison between a centroid-based method and a density-based method. This comparison provides valuable insights into how different clustering techniques perform on the same dataset allowing the strengths and limitations of each algorithm to be evaluated. Furthermore, using both algorithms helps determine which clustering approach is more suitable for identifying hidden patterns and grouping similar individuals.

4.3 Development of the Clustering Models

4.3.1 K-Means

The K-Means clustering model was developed using the PCA-transformed dataset. Since the optimal number of clusters was not known beforehand, an automatic search was conducted by testing different values of K ranging from 2 to 9. For each tested K value, a K-Means model was trained, cluster assignments were generated, and the Silhouette Score was calculated to evaluate the quality of the clustering result. The K-Means algorithm was implemented using random_state=42 to ensure reproducible results and n_init=10 to perform multiple centroid initializations for improved stability. The K value that achieved the highest Silhouette Score was selected as the optimal number of clusters. After determining the best K value, the final K-Means model was trained and additional clustering information, including cluster centroids, cluster size distribution, inertia value, and Silhouette Score, was obtained for further analysis.

4.3.2 DBSCAN

The DBSCAN clustering model was developed using the same PCA-transformed dataset. Unlike K-Means, DBSCAN does not require the number of clusters to be defined before training. Instead, it identifies clusters based on the density distribution of data points using two main parameters which are eps and min_samples. To obtain the optimal DBSCAN configuration, multiple combinations of parameters were tested. The eps value was evaluated within the range 0.10 to 0.95, while min_samples values between 3 and 9 were considered. Each parameter combination was evaluated using the Silhouette Score after excluding noise points labelled as -1. The parameter combination that produced the highest Silhouette Score was selected as the final DBSCAN model. The final model was then used to determine the number of clusters, identify noise points, and analyse the distribution of samples within each cluster.

4.4 Determination of the Number of Clusters

4.4.1 K-Means

K-Means requires the number of clusters to be predefined. Therefore, different K values ranging from 2 to 9 were tested, and the Silhouette Score was calculated for each value. A higher Silhouette Score indicated better cluster separation and stronger similarity within cluster. The highest Silhouette Score was obtained at K = 3 with a score of 0.8795. Therefore, three clusters were selected as the optimal number of clusters for the K-Means model.

4.4.2 DBSCAN

DBSCAN determines the number of clusters automatically based on the density distribution of the dataset. After performing parameter optimization, the best DBSCAN configuration was obtained using eps = 0.10 and min_samples = 9. This configuration produced a Silhouette Score of 0.8851 and identified 3 clusters. Additionally, DBSCAN detected 45 noise points, representing approximately 3.8% of the dataset, which were considered observations that did not belong to any dense cluster.

4.5 Results Produced by Each Clustering Algorithm

Table 4.5.1 shows the K-Means algorithm successfully divided the dataset into three distinct clusters using the optimal K value of 3. The resulting clusters were relatively balanced, consisting of 411 samples (34.2%), 391 samples (32.6%), and 398(33.2%). The Silhouette Score of 0.8795 indicates that the clusters were well separated with strong similarity among data points within the same clusters.

Similarly, the DBSCAN algorithm identified three density-based clusters using the optimized parameters of eps = 0.10 and min_samples = 0. The model achieved a slightly higher Silhouette Score of 0.885q compared to K-Means. Furthermore, DBSCAN detected 45 noise points (3.8%) , demonstrating its capability to identify unusual observations or outliers that may not belong to any cluster. This highlights an advantage of DBSCAN compared to K-Means, as K-Means assigns every data point to a cluster and does not explicitly detect noise.

Model	**Parameters**	**Silhouette Score**	**Number of Clusters**	**Noise Points**
K-Means	K = 3	0.8795	3	0
DBSCAN	eps = 0.10, min_samples = 9	0.8851	3	45

Table 4.5.1 : Summary of Clustering Model Results

5.0 Model Performance and Evaluation

This section evaluates the performance of the two clustering models developed in this project that are K-Means and DBSCAN. Since clustering is an unsupervised learning task, the evaluation relies on internal validation metrics that measure the quality of the clusters based on the data's inherent structure rather than external labels. The primary metric used is the Silhouette Score, which provides a measure of how similar a data point is to its own cluster compared to other clusters.

5.1 Performance Analysis of K‑Means Clustering

The K-Means algorithm was applied to the dataset using the optimal number of clusters, k=3, which was determined based on the highest Silhouette Score across a range of K values from 2 to 9.

The K-Means model achieved a Silhouette Score of 0.8795. This is a very high score, indicating that the three clusters are exceptionally well-separated and internally cohesive. The proximity of this score to +1.0 suggests that the algorithm successfully identified distinct groupings of teenagers based on their behavioral and psychological features.

The algorithm produced three relatively balanced clusters, containing 411 (34.2%), 391 (32.6%), and 398 (33.2%) samples. This balance is a positive indicator, as it suggests the model did not create a dominant, overpopulated cluster that would dilute the analysis.

While not explicitly computed in the provided results, the inertia is a key metric for K-Means. A lower inertia generally indicates a better model. The high Silhouette Score and the balanced cluster sizes collectively imply that the model converged to a stable and meaningful solution.

The high performance of the K-Means model is also supported by the data preprocessing steps. The application of StandardScaler for normalization ensured that features with larger numerical ranges did not dominate the distance-based calculations. The removal of the depression_label successfully prevented target leakage, allowing the algorithm to form natural, behavioral-based clusters rather than ones biased by the outcome variable.

5.2 Performance Analysis of DBSCAN Clustering

The DBSCAN algorithm was applied with optimized (eps=10, min_samples=9) that were selected to maximize the Silhouette Score.

The DBSCAN model achieved a slightly higher Silhouette Score of 0.8851, outperforming the K-Means model. This marginal improvement suggests that the density-based approach was slightly better at capturing the natural grouping structure of the data.
DBSCAN identified 3 clusters and also flagged 45 data points (3.8%) as noise. These noise points are observations that did not belong to any dense region. This is a significant strength of the DBSCAN algorithm, unlike K-Means, which forces every data point into a cluster, DBSCAN can identify outliers or unique cases that do not fit any profile. In a mental health context, these 45 individuals might represent unique or edge cases that require special attention.

The effectiveness of DBSCAN relies heavily on proper feature scaling, as it uses Euclidean distance to build its density-based neighborhoods. The standardization step was crucial to ensure that no single feature artificially defined the neighborhood radius (eps). The removal of the depression_label also ensured that clusters were formed based on the independent behavioral and usage features.

5.3 Comparison of K-Means and DBSCAN

The comparison between K-Means and DBSCAN reveals important differences in their approach, performance, and practical applicability for analyzing the Teen Mental Health dataset.

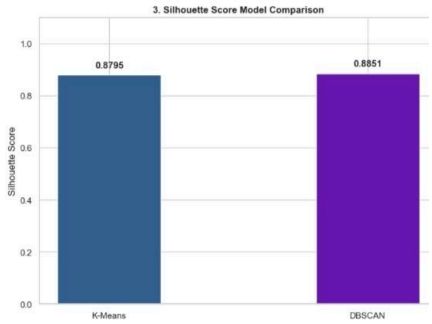


Figure 5: Silhouette Score Mod Comparison

Table 5.1: Comparison of K-Means and DBSCAN Clustering Algorithms

Feature	K-Means	DBSCAN
Clustering Approach	Partitioning‑based (centroid)	Density-based
Silhouette Score	0.8795	0.8851
Number of Clusters	3	3
Outlier Detection	No (assigns all points)	Yes (identifies 45 noise points)
Interpretability	High	Moderate

Strengths	Simple, efficient, produces clear centroids for profile definition.	Can find clusters of arbitrary shape and handles outliers effectively.
Weaknesses	Assumes spherical clusters and is sensitive to outliers.	Requires careful parameter tuning.

As shown in Table 5.1, both K-Means and DBSCAN identified three clusters in the Teen Mental Health dataset, with DBSCAN achieving a marginally higher Silhouette Score of 0.8851 compared to K-Means at 0.8795. This slight improvement suggests that the density-based approach of DBSCAN was better at capturing the natural grouping structure of the data. However, the key difference between the two algorithms lies in their handling of outliers. DBSCAN successfully identified 45 noise points (3.8% of the dataset) that did not belong to any cluster, demonstrating its advantage in detecting unusual observations that may require special attention in a mental health context. In contrast, K-Means assigned every data point to a cluster, which may potentially mask unique cases that do not fit typical profiles.

K-Means offers greater interpretability due to its clear cluster centroids, making it easier to define distinct profiles. On the other hand, DBSCAN's ability to handle irregular cluster shapes and detect outliers provides a more robust analysis, though it requires careful parameter tuning. The complementary strengths of both algorithms suggest that using them together offers a comprehensive understanding of the data, where K-Means helps identify standard student profiles while DBSCAN highlights exceptional cases requiring personalized intervention.

6.0 Visualization and Discussion of Results

This section visualizes the clustering results generated by the K-Means and DBSCAN algorithms to facilitate interpretation and discussion.

6.1 Visualization of Clustering Results

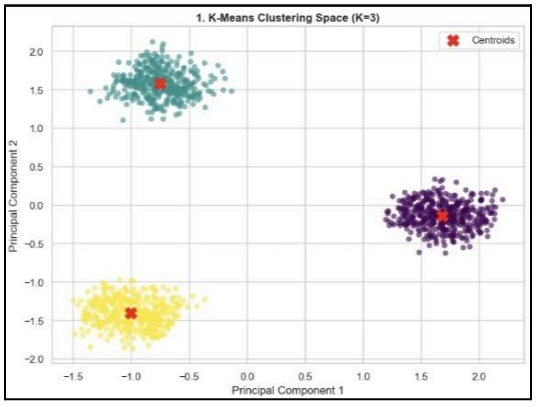


Figure 6.1.1: K-Means Clustering Visualization (PCA-Transformed Data)

The scatter plot shows the data points projected onto the first two Principal Components (PC1 and PC2). The points are colored according to their assigned K-Means clusters. Cluster 0 (Purple) represents 411 data points, primarily located in the upper-right and lower-right quadrants of the plot. Cluster 1 (Cyan) represents 391 data points, concentrated in the upper-left quadrant. Cluster 2 (Yellow) represents 398 data points, predominantly clustered in the lower-left and lower-center regions.

The three clusters appear distinctly separated, with relatively little overlap between them, visually confirming the high Silhouette Score of 0.8795.

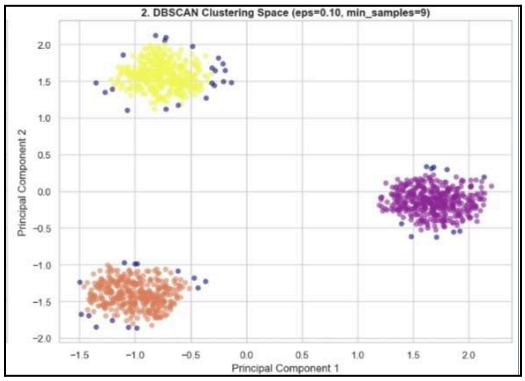


Figure 6.1.2: DBSCAN Clustering Visualization (PCA-Transformed Data)

The scatter plot visualizes the DBSCAN clustering results of Cluster 0, 1, 2. The three identified clusters occupy regions of high density. The 45 noise points are scattered sporadically across the plot and do not fall within the dense neighborhoods of any cluster.

The DBSCAN visualization shows clusters with slightly more complex and irregular shapes compared to K-Means, which fits the algorithm's density-based nature. The presence of noise points scattered across the plot indicates that DBSCAN can successfully identify and separate outliers, providing a more robust visualization of the data's natural structure.

6.2 Explanation of Cluster Characteristics

By examining the feature values (centroids for K-Means and density distributions for DBSCAN) associated with each cluster, we can develop clear profiles of the groups. Based on the K-Means centroids (which are easier to interpret)

The "High-Risk Digital Dependent" cluster comprises 411 students whose profiles reveal a strong correlation between a heavy digital lifestyle and poor mental well-being. This group is characterized by the highest daily social media hours, the most screen time before sleep, and the lowest levels of physical activity among all participants. Consequently, these individuals report

high levels of stress, anxiety, and digital addiction. The data suggests that they spend excessive time online, likely at the direct expense of their sleep quality and physical health, which in turn exacerbates their psychological distress.

The "High-Performing, Socially Active" cluster consists of 391 students who demonstrate that moderate digital usage does not negatively impact overall well-being. This group stands out by maintaining the highest social interaction levels and the best academic performance, all while keeping their screen time moderate. As a result, these individuals report moderate to low levels of stress, anxiety, and digital addiction. Ultimately, this profile suggests that a balanced lifestyle anchored by robust real-world social engagement and a strong academic focus acts as a vital protective factor against mental health risks.

The "Moderate, Balanced" cluster comprises 398 students who represent the average teenager, maintaining a relatively stable lifestyle without excelling or struggling in any particular area. This group exhibits moderate values across all measured features, including balanced sleep patterns, moderate social media usage, and average academic performance. Reflecting this stability, their mental health metrics also fall squarely in the moderate range. While these individuals are not at immediate risk, this profile serves as an ideal baseline group for comparison against more extreme clusters.

6.3 Discussion of Important Patterns

The strongest takeaway from the results is the protective role of a balanced lifestyle. Students in Cluster 1, who maintain high social interactions and good academic performance alongside moderate digital usage, are less likely to exhibit signs of stress and anxiety. This pattern suggests that interventions should not solely focus on reducing screen time but on promoting holistic health by encouraging physical activity, social connectedness, and good sleep hygiene.

The clustering approach successfully categorizes teenagers into risk groups without needing a medical diagnosis. This allows for proactive, rather than reactive, mental health support. The identification of the "High-Risk Digital Dependent" group (Cluster 0) enables schools to implement early intervention programs for students who might not have yet sought help but are displaying clear risk factors.

The presence of outliers in the DBSCAN model (3.8% of the population) serves as a powerful reminder that general profiles are not universal. For these individuals, standard interventions may be ineffective. This underscores the need for personalized mental health assessments to complement the broad, data-driven profiles discovered through clustering.

7.0 Conclusion

In conclusion, this project has effectively demonstrated the effectiveness of unsupervised machine learning in discovering hidden structure and meaningful patterns within the dataset. Through applying K-Means and DBSCAN clustering algorithms to the Teen Mental Health dataset, future decisions could be made through group clustering. Combining data preprocessing, feature standardization, and dimensionality reduction via PCA identifies the distinct behavioral clusters, proving that complex interactions between digital habits, physical activities, and psychological metrics without having any pre-diagnosis label attached to a teenager.

These results prove to be very useful in creating an empirical basis of profiles of those young people who are at risk of developing depression based on their habits. The specific combinations of screen time, sleep quality, and social interaction patterns naturally group together, educators and mental health professionals can implement more effective strategies to prevent depression and maintain good mental health among youths. This project confirms that unsupervised machine learning is a powerful tool in addressing the growing mental health challenges faced among teenagers, offering a method to understand and support youth well-being in an increasingly digital world.

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